Senith Jayakody

AI Research and Development Engineer – FARBE Technologies

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Research Interests

My research interests include **signal processing**, **machine learning and statistical learning**, and **image processing**, with applications in biomedical signal processing, medical imaging, and computational biology.

Education

B.Sc. Eng. (Hons.) in Electrical and Electronic Engineering *2021 – 2025*
University of Peradeniya, Sri Lanka GPA: **3.85/4.00**

First Class Honours | Class Rank: **6/115** | Academic Transcript

G.C.E. Advanced Level Examination *2019*
Combined Mathematics (A), Physics (A), Chemistry (A)

Result Sheet

Publications

Peer-Reviewed Journal Articles

[J1] Ranasinghe, G. A. S. L.[†], **Jayakody, J. A. S. T.**[†], De Silva, M. C. L., Thilakarathne, G., Godaliyadda, G. M. R. I., Herath, H. M. V. R., Ekanayake, M. P. B., and Navaratnarajah, S. K., "Reflectance multispectral imaging for soil grain-size composition estimation and USDA texture classification," *IEEE Access*, 2026. DOI: 10.1109/ACCESS.2026.3721276

Manuscripts Under Review

[M1] Tilakarathna, U.[†], **Jayakody, S.**[†], Jayasooriya, K., Godaliyadda, R., Ekanayake, M. P., Nawinne, I., and Rathnayake, C., "Empirical mode decomposition and interpretable machine learning for preterm birth classification from electrohysterography," Manuscript under review at *PLOS ONE*, 2026. DOI: 10.48550/arXiv.2608.17643

[†] The first two authors contributed equally to [J1] and [M1].

Conference Papers Under Review

[C1] Apakanda, B., Ranasinghe, S., **Jayakody, S.**, Appuhami, H. D. M. D., Godaliyadda, R., Ekanayake, M. P., and Ragel, R., "Colour-INR: Implicit neural representation for image colorization through colour guidance," Under review at the 15th International Conference on Learning Representations (ICLR 2027), 2026

Peer-Reviewed Conference Papers

[C2] Perera, S., Ranasinghe, S., **Jayakody, S.**, Epakanda, B., Godaliyadda, R., and Ekanayake, M. P., "Feature conditioned diffusion for audio generation," in *2025 IEEE 19th International Conference on Industrial and Information Systems (ICIIS)*, 2026, pp. 162–167. DOI: 10.1109/ICIIS69028.2026.11450741

[C3] Ranasinghe, S., **Jayakody, S.**, Silva, M. D., Herath, V., Godaliyadda, R., Ekanayake, M. P., Navaratnarajah, S., Kizel, F., and Thilakarathne, G., "Deep learning for soil moisture content estimation via reflectance multispectral imaging," in *2024 International Conference on Advances in Technology and Computing (ICATC)*, 2024, pp. 1–6. DOI: 10.1109/ICATC64549.2024.11025290

Honors and Awards

Second Runner-Up, Manamperi Award – Recognized in the Best Undergraduate Project in Sri Lanka competition for the final-year research project "Multispectral Imaging" *2025*

Silver Award, Engineering Category – Sahasak Nimavum Inventions and Innovations Competition, for the final-year research project "Multispectral Imaging" *2024*

Top 10 Finalist, Predicta v1.0 – Organized by the IEEE Student Branch, University of Peradeniya *2024*

Finalist, IEEE Electronic Design Competition – Organized by the IEEE Sri Lanka Section *2023*

Second Place, ACES Hackathon – Healthcare Category; organized by the Association of Computer Engineering Students, University of Peradeniya *2023*

Fourth Place, IESL RoboGames – University Category, Institution of Engineers, Sri Lanka *2022*

Selected Research Projects

Preterm Birth Prediction from EHG and Tocographic Signals *2025 – Present*

Developing signal-processing and machine-learning approaches for preterm-birth prediction using electrohysterography (EHG) and tocographic signals.

- Developed spectrotemporal feature-extraction pipelines using spectral, wavelet, entropy-based, and statistical signal characteristics.
- Investigated empirical mode decomposition (EMD) for extracting interpretable features from uterine electrophysiological signals.

Concepts: Biomedical Signal Processing, EHG, Tocography, Time-Frequency Analysis, EMD, Feature Extraction, Machine Learning.

Image Colorization using Implicit Neural Representations *2026 – Present*

Investigating image colorization using implicit neural representations (INRs), where continuous coordinate-based image representations are guided by sparse colour information.

- Formulated image colorization as an inverse problem using coordinate-based neural representations.
- Investigated the integration of sparse colour guidance for controlling the reconstructed colour distribution while preserving image structure.

Concepts: Implicit Neural Representations, Image Colorization, Inverse Problems, Computational Imaging, Deep Learning.

Theoretical Analysis of Diffusion Models *2026 – Present*

Investigating the statistical and theoretical behaviour of denoising diffusion models, with emphasis on the properties of predicted noise and their relationship to generative fidelity.

- Studied systematic differences between the statistical properties of true diffusion noise and model-predicted noise during the reverse process.
- Investigated trace-energy behaviour and regularization strategies for improving the fidelity of generated samples.

Concepts: Denoising Diffusion Probabilistic Models, Generative Modeling, Statistical Learning, Noise Prediction, Covariance Structure, Regularization.

Multispectral Imaging for Environmental and Biological Sensing *2023 – 2025*

Developed multispectral imaging systems and machine-learning methods for non-destructive characterization of environmental, agricultural, and biological samples.

- **Soil Analysis:** Developed reflectance multispectral imaging methods for soil-moisture estimation, grain-size composition prediction, and USDA texture classification.
- ***Microcystis* and Microcystin Estimation:** Developed transmission-mode multispectral imaging methods for estimating *Microcystis* cell concentration and ELISA-equivalent microcystin concentration.
- **Imaging-System Development:** Contributed to the design and development of custom multispectral imaging hardware, including wavelength-controlled illumination, embedded control, image acquisition, calibration, and preprocessing.

Concepts: Multispectral Imaging, Spectral Analysis, Computational Imaging, Machine Learning, Regression, Classification, Embedded Sensing.

Research and Professional Experience

AI Research and Development Engineer *Aug 2026 – Present*
FARBE Technologies (Pvt) Ltd Sri Lanka

- Developing AI-based signal and image processing solutions for research and industrial applications.

Research Assistant *Aug 2025 – Jul 2026*
Multidisciplinary AI Research Centre (MARC), University of Peradeniya Peradeniya, Sri Lanka

- Conducted research in biomedical signal processing, generative models, machine learning, and multispectral imaging.
- Guided junior researchers and contributed to experimental planning, data analysis, and manuscript preparation.

Research and Development Intern *Jul 2024 – Oct 2024*
Swartup Pvt. Ltd. Sri Lanka

- Worked on embedded-system development using nRF5340, Zephyr, LoRa, and GSM technologies.

Research and Development Intern *Jul 2023 – Oct 2023*
SLT Digital Lab, Sri Lanka Telecom PLC Sri Lanka

- Developed IoT-based environmental monitoring systems using ESP32, MQTT, Firebase, and integrated sensors.

Technical Skills

- **Programming:** Python, MATLAB, C/C++, Assembly
- **Machine Learning and Deep Learning:** PyTorch, TensorFlow, scikit-learn; supervised learning, generative modeling, model evaluation and validation
- **Signal and Image Processing:** Spectral and time-frequency analysis, wavelet analysis, empirical mode decomposition (EMD), MFCC, OpenCV, SciPy
- **Embedded and Sensing Systems:** Raspberry Pi, ESP32, nRF5340, Zephyr, FreeRTOS, sensor integration, circuit design and calibration
- **Research and Development Tools:** Linux, Git, LaTeX, Altium Designer, EasyEDA

Academic Service and Outreach

Workshop Facilitator – Conducted AI and embedded-systems sessions for pre-engineering students at the Diamond Jubilee AI Workshop *2025*

Research Demonstrator and Speaker – Presented real-world AI and multispectral-imaging applications at the Trinity College AI Symposium and DreamEx *2025*

Logistics Team Member – Supported robotics competitions and technical workshops, IEEE Robotics and Automation Society Student Chapter *2021 – 2022*

Volunteer – Supported technical sessions and student-engagement activities, IET on Campus *2022 – 2023*

Academic References

1. **Prof. G. M. R. I. Godaliyadda**
   Ph.D., National University of Singapore; B.Sc. Eng. (Hons.), University of Peradeniya
   Professor, Department of Electrical and Electronic Engineering, University of Peradeniya
   roshang@eng.pdn.ac.lk | Office: +94 81 239 3431 | Mobile: +94 77 770 9035

2. **Prof. H. M. V. R. Herath**
   Dr.-Ing., University of Paderborn, Germany; M.S., University of Miami, USA; B.Sc. Eng., University of Peradeniya
   Professor, Department of Electrical and Electronic Engineering, University of Peradeniya
   vijitha@eng.pdn.ac.lk | Office: +94 81 239 3436 | Mobile: +94 71 673 2688

3. **Prof. M. P. B. Ekanayake**
   Ph.D. in Applied Mathematics, Texas Tech University, USA; B.Sc. Eng. (Hons.), University of Peradeniya
   Professor, Department of Electrical and Electronic Engineering, University of Peradeniya
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